

License Plate Recognition

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Project Overview

The goal of this project was to develop a complete automatic license plate recognition (ALPR) pipeline using only OpenCV and classical computer vision techniques. The system was designed to detect a vehicle license plate from an image, isolate the characters on the plate, and recognize the alphanumeric sequence using machine learning based optical character recognition (OCR).

Unlike modern deep learning based systems, this project intentionally avoided external OCR libraries and neural networks in order to better understand the underlying computer vision and machine learning concepts involved in license plate recognition.

The final system was implemented in C++ using OpenCV and consisted of four major stages:

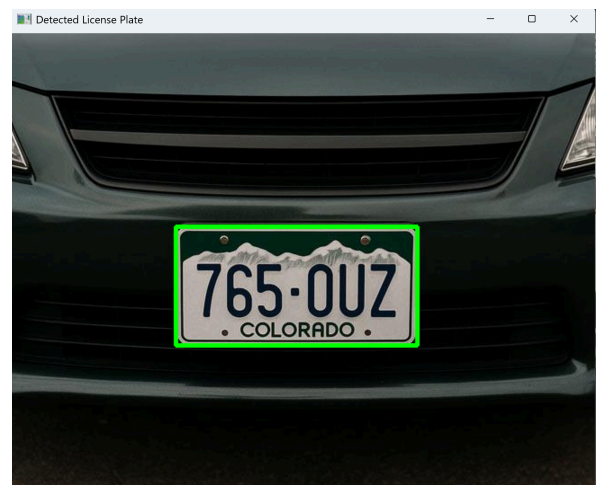
1. License plate detection
2. Text region extraction
3. Character segmentation
4. Character recognition using HOG and SVM

System Implementation

1. License Plate Detection

The first stage of the pipeline identifies the most likely license plate region in the input image. This process involved:

- Image preprocessing
- Edge detection
- Contour extraction
- Aspect ratio filtering
- Rotated rectangle analysis



Candidate regions were evaluated using geometric constraints commonly associated with license plates, such as: width to height ratio, rectangular shape, and contour area. The best candidate was extracted and perspective corrected for further processing.

2. Text Detection and Character Segmentation

After extracting the plate image, the system isolated the primary text region while ignoring:

- Plate borders
- Decorative graphics
- State names
- Screws and mounting hardware



The plate image was converted to grayscale and thresholded using Otsu thresholding to create a binary image. Morphological operations were applied to reduce noise, separate touching regions, and improve contour stability.

Contours were then analyzed to locate possible characters. Each contour was filtered using:

- Relative height
- Relative width
- Aspect ratio
- Area constraints

The remaining valid contours were sorted left to right and cropped into separate character images for OCR recognition.

3. Text Recognition

Character recognition was implemented using Histogram of Oriented Gradients (HOG) features combined with a Support Vector Machine (SVM) classifier.

HOG Feature Extraction

Each segmented character image was:

- Normalized to a fixed size
- Centered within the image
- Converted into gradient based feature descriptors

Afterwards, HOG captures: stroke direction, edge structure, and character shape information to create a compact numerical representation of each character.

SVM Classification

The HOG feature vectors were used to train a multiclass Support Vector Machine classifier. The classifier learned decision boundaries between letters A-Z and digits 0-9.

Synthetic training templates were generated using randomized transformations including: rotation, translation, blur, dilation, erosion, and noise injection. This significantly improved recognition robustness.

Results

The final system successfully performed plate localization, character segmentation and character recognition. However, the most difficult portion of the project was OCR accuracy. Early versions of the classifier incorrectly predicted nearly all characters due to:

- Poor template diversity
- Inconsistent image polarity
- Improper character normalization

After improving preprocessing and template generation, recognition accuracy improved substantially.

Table 1: Example Recognition Progress For Plate 7650UZ

Version	Example Output
Initial OCR	999999
Improved Preprocessing	11EIIZ
Improved Templates	765IIZ
Final System	7650UZ

While the system still struggled with visually similar characters such as O and 0, the overall recognition quality improved significantly over the course of development.

Additional Challenges Encountered

We encountered several other technical challenges during the project such as:

- **Similar character shapes:** Characters with similar structures sometimes confused the classifier such as O and 0 or l and 1.

- **Thresholding variability:** Lighting conditions and reflections caused inconsistent binary images.
- **Contour filtering:** Small changes in filtering thresholds could remove valid characters, merge characters together, or detect false positives.
- **Training performance:** Increasing HOG feature resolution dramatically increased SVM training time and memory requirements.

Lessons Learned

This project provided valuable experience with classical computer vision and machine learning techniques. It also demonstrated the tradeoffs between accuracy, computational cost, and simplicity of implementation.

Key lessons from this project include:

- OCR accuracy depends heavily on preprocessing quality
- Small preprocessing inconsistencies can heavily impact recognition performance
- Character normalization is critical for reliable feature extraction
- HOG features remain highly effective for shape based recognition tasks
- Traditional computer vision pipelines require careful tuning but provide strong interpretability

Conclusion

This project successfully implemented a complete license plate recognition system using only OpenCV and classical machine learning techniques. The system was capable of:

- Detecting license plates
- Extracting text regions
- Segmenting characters
- Performing OCR recognition

Although the final OCR accuracy was not perfect, the project achieved substantial improvements through iterative preprocessing, normalization, and template generation refinements.